

Building accurate legal case outcome prediction models.

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Abstract: The legal community is very interested in and highly valued by legal predictive models. Lawyers and judges in particular can use these models' current benefits to forecast case difficulties to further complicate their future courses of action, such as faster decision-making, providing evidence for arguments, supporting the opposition, etc.. Yet it takes a long time and involves a lot of complex stages to correctly predict court rulings and case issues, including data mining, sanctification, engineering, and the chance discovery of appropriate bulk case papers. However, the legal complexities make everything more challenging. Here used several machine learning algorithms to predict the legal case. Naïve Bayes, Logistic Regression and Gradient Boosting algorithms are used. Here the accuracy and the confusion matrix of classifiers is evaluated.

Keywords—Data Mining, Machine Learning, Legal cases, Case Prediction, Datasets, Algorithm.

I. INTRODUCTION

In the 21st era, technological advances have affected a lot of aspects of our everyday lives. Imagine a situation where all legal decisions are made by machines rather than judges. That sounds challenging, doesn't it? If you look at the scenario differently, it actually has a benefit because it will result in objective assessments. Currently, before presenting their case's conclusion to the judge, the majority of attorneys personally assemble facts and compare it to previous court rulings and statutes. The decision will be made after manually analyzing the supplied data. Daily basis, they have more information to review. It will take a while for the verdicts for each case to be rendered. Due to faulty judgement caused by corruption, Indians' confidence in the judicial process is very less. In India, numerous instances of biased judgement have been documented.[5]

The political parties in power will also have an impact on the judgement, which frequently results in wrong and unlawful decisions. In order to increase judicial uniformity, justice for all, and administrative effectiveness, predictions of judgements have long been a focus of discussion in legal theory and in practice. As a result, a variety of approaches and methodologies, from straightforward mathematical models to

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extremely sophisticated analytical algorithms, have been established over the years to forecast legal decisions. There have been many different methods tried; in particular, with the recent development of AI, methods based on artificial intelligence (AI) have been used more frequently. In order to assist lawyers, artificial intelligence (AI) algorithms are being built that forecast verdicts easily and quickly.[2]

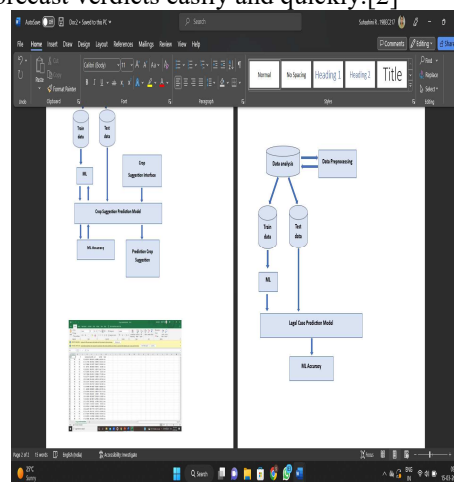


Fig 1. System architecture for legal case outcome prediction

The prospects of adopting automatic data extraction and natural language processing to facilitate statistical tests in the legal system is discussed. More precisely, it is established and explained how NLP methods might be utilised to anticipate court judgements made instantly by Indian courts. Predicting case verdicts is the major objective of this research endeavour. This makes acquiring sufficient legal data the most important component of the research project. Using a natural language processing technology, keywords will be extracted from the data after collection. Dividing the supplied text into various categories by means of neural networks.

II. LITERATURE SURVEY

Josh Blackman, Michael et al.[3] In order to facilitate the effective employing technology to process legal processes and resources, the emerging field of judicial informatics is penetrating the current, complicated legal system. The consequence has been impossibly slow and time-consuming because the legal system generates large amounts of textual material through procedures, opinions, legislation, judgements, legal contracts, etc. Because of their intricate nature and format, legal texts and documents must undergo a thorough employing technology to process legal processes and resources,

In[4],In this study, the Delhi District Court's decisions are used to construct a model to forecast the results of murder-related cases. Classification using machine learning methods were utilized to predict 'Acquittal' or 'Conviction' of suspect after locating and assembling essential legal components for 86 murder connected cases. The outcomes are successfully predicted, and the findings show the applicability of the criteria taken into consideration. Yet, There seems to be a defect, specifically, poor performance in anticipating outcomes for situations with several accusers and outcomes that differ for different accusers.

Kevin D et al[7], This study describes, Since many years, the Indian legal system relied on the physical gathering and processing of evidence and information for judgements; but, with the introduction of our CNN model, accurate and impartial judgements may be rendered in a significantly shorter amount of time. It will also be very beneficial to promote less corruption and impartial judgements. Study of this paper, illustrates the value of natural language processing (NLP) and convolutional neural networks (CNN) in the judicial decision-making process by extracting words from the court data and utilising CNN to accurately predict the cases' outcomes with 85% precision. The results show that the model is more efficient and that the case only carries one charge. The ability to accurately forecast the outcome declines with the amount of charges.

III. DATASETS

The dataset consists of attributes such as IPC section, Evidence, Label and Punishment. We have created the dataset with the reference of several datasets collect from online resources. Enormous number of instances have been collected from the past historical cases. This dataset comprise number of cases such as murder, theft, drugs, accident, fraud, kidnapping, sexual assault, rape, smuggling, cheating, false complaint, vandalism, blackmail, Driving without license, embezzlement, trespassing, traffic violation, mischief, Wild life protection, Harassment and money laundering ,...etc. Several kinds of cases represented by the label are demonstrated below.

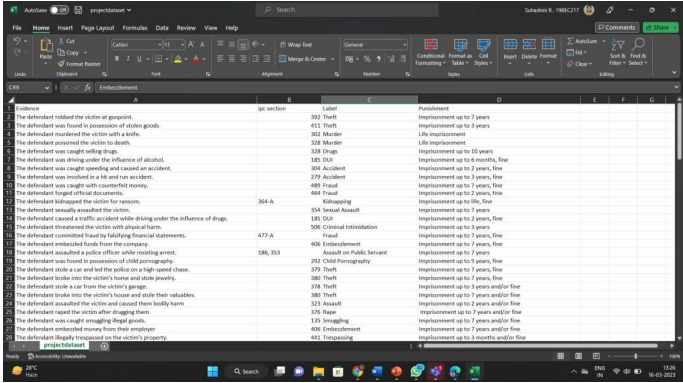


Fig 2. Dataset

Title	Number of cases
Accident	89
Assault	75
Blackmail	81
Murder	97
Corruption	84
Theft	96
Cybercrime	90
Sexual Assault	73
Cheating	85
Smuggling	79
Drugs	93
Kidnapping	75

Table 1. Case count

IV. METHODOLOGY

The suggested technique's process starts with gathering information, undergoes certain pre-processing, and ends with analysing the information. data is split into training data and test data and the selection of model and the model implementation and finally it predicts the verdict.

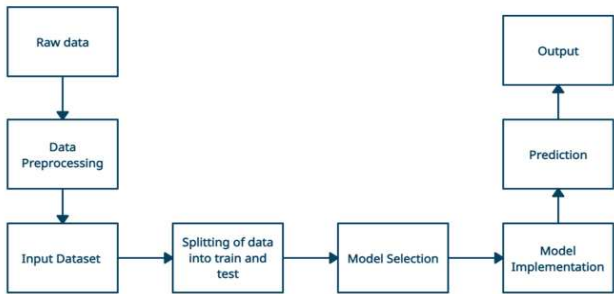


Figure 1. Block Diagram of Crop recommendation Model

Fig 3. Process flow model for automation

A. Dataset Collection

The review of the information is one of our first execution procedures. To determine whether there were any relationships

The dataset contains parameters such as murder, theft, drugs, accident, fraud, kidnapping, sexual assault, rape, smuggling, cheating, false complaint, vandalism, blackmail, Driving without license, embezzlement, trespassing, traffic violation, mischief, Wild life protection, Harassment and money laundering etc.

The datasets have been created by us with reference of some dataset from online resources. The information collection contains a number of occurrences that were obtained from past records.[15]

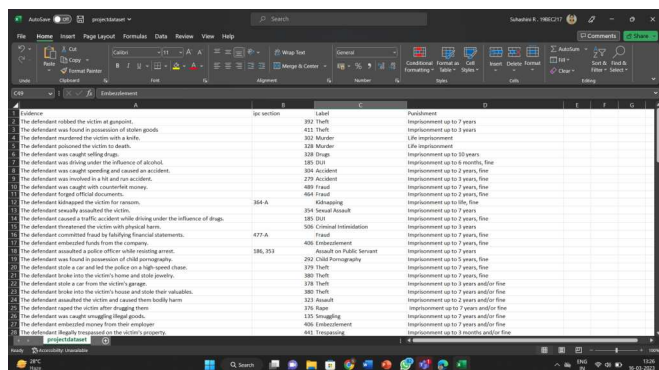


Fig 4. Collection of Dataset

B. Dataflow Representation

A dataflow image is sometimes known as a bubble chart. This straightforward visual representation can be used to illustrate the information that is provided to the system, the several processes that are carried out on that data, and the output data that the system produces.[10]

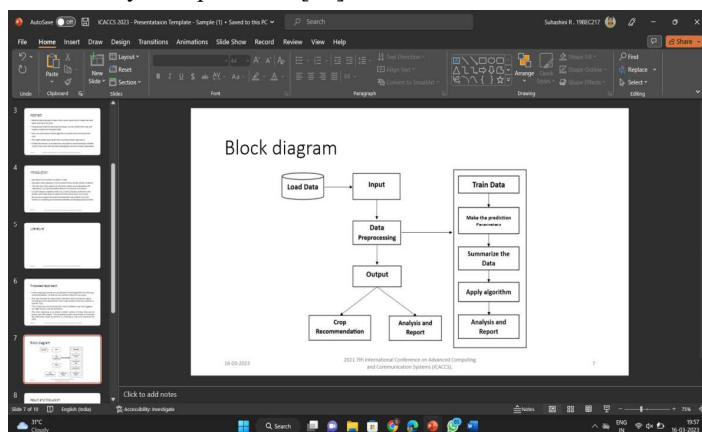


Fig 5. Process flowchart

C. Training and Extraction Phase

The characteristic is obtained after pre-processing a batch of data from the last twenty years. The following is a summary of the learning phase:

- To collect characteristics such as case id, IPC section, Label and Evidence.
- These characteristics are employed while training a machine learning classifier.

The final outcome of the training process is the learned algorithm, which can suggest instances based on the applicable IPC section and the supporting data. A trained classifier's efficiency is assessed using the metrics. The actions are Accuracy and confusion matrix.[9]

D. Classification

Following is an expression for the categorization period:

- The retrieved variables are used as input data.
- A trained classification recommends trimming based on input variables.

This phase's result is the accuracy and confusion matrix according to the given input.

D. Preprocess the Dataset file

Select the source information collection, select the attachment using the "Upload" option, and pre-process the data. The dataset should be uploaded after selecting the open file option in the pre-process tag.

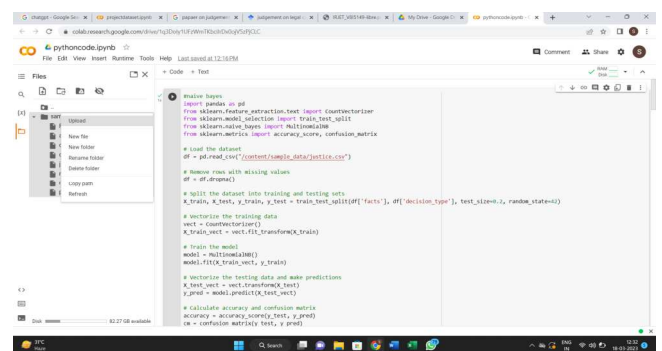


Fig 6. Creation of data set file

V. MACHINE LEARNING ALGORITHMS:

There are several machine learning algorithms. Here, we used naïve bayes, Gradient Boosting and Logistic Regression algorithms to calculate the accuracy and the confusion matrix.

A. Naïve Bayes algorithm:

The "naive" word from Naive Bayes originates from the idea that the input features are unrelated to one another. in spite of this reduced complexity, Naive Bayes is capable of excellent performance in real-world settings, particularly when the input features are not highly correlated and when there is a significant quantity of training information available.[16]

The quickness and effectiveness of Naive Bayes is one of its main benefits. The technique is well-suited for real-time applications or huge datasets because it just needs a small amount of training data and can be taught quickly.[8]

However, in circumstances when the input characteristics are highly correlated or when there are complex correlations between the input features and the target features, Naive Bayes may not perform as well as more complex algorithms. In addition, Naive Bayes makes the assumption that the input features are independent, which may not be accurate in many situations in everyday life.

B. Gradient Boosting algorithm:

Gradient Boosting is a machine learning approach that creates a group of ineffective predictors, typically decision trees. The system operates by constantly adding additional decision tree structures to the approach, with each one aiming to fix the mistakes produced by the ones before it.[13]

The method begins training with adapting a straightforward decision tree to the data. The variance among the initial tree's projections and the actual target values is then calculated. It trains a second tree using this error as a target variable. The following additional tree is constructed to foretell the discrepancy between the actual target values and the first tree's forecasts. Each successive tree in this procedure is focused on fixing the mistakes produced by the prior trees.

The final outcome is created by combining all of the tree' forecasts. This method produces a model that is resistant to overfitting and capable of being highly precise across a variety of tasks.[12]

C. Logistic Regression algorithm:

In binary categorization assignments, where the objective is to forecast a binary output factor (for example, yes/no, true/false) based on a set of input variables or characteristics, the commonly used machine learning method logistic regression is utilized.

Modelling the relationship between the input variables and the likelihood that the output variable will be 1 is how the algorithm operates. The result of a linear regression analysis is converted into a probability value between 0 and 1 using a logistical function, often known as the sigmoid function.[14]

By reducing the gap that exists between anticipated probabilities and the actual binary outcomes in the training data, the method determines the logistic perform variables while learning. Finding the combination of variables that optimizes the likelihood of the observed data is the goal.

Once instructed, the algorithm can be used to make predictions based on fresh data. The algorithm uses the logistical function to determine the likelihood that an output variable will equal 1 given a collection of input values. The result is forecasted to be 1 if the probability is greater than a predetermined cut-off (often 0.5). If not, it is anticipated to be 0.[17]

VI. RESULTS AND ANALYSIS

After the model is trained, the accuracy and the confusion matrix is displayed for each algorithm.

A. Accuracy:

Precision is more lucid. It determines how well our performance predictions by comparing our show predictions with the exact proportion figures.[1]

B. Confusion Matrix:

The display of the surface names versus planer using Confusion Network may be unbelievable. Each row in the matrix refers to incidences in a real lesson, and each push refers to instances in a forecasted lesson. Confusion Matrix isn't really an execution meter, but it serves as a foundation for other measurements to evaluate outcomes.[6]

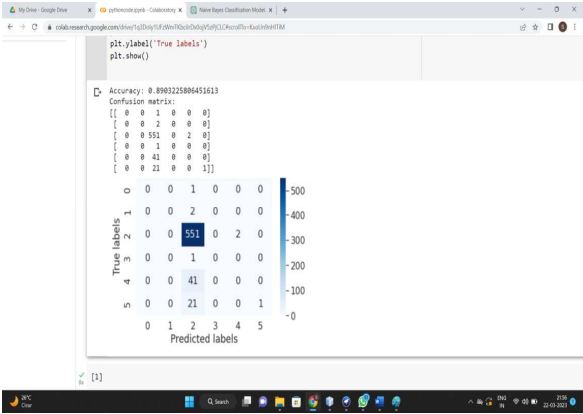


Fig.8 Naive Bayes Algorithm

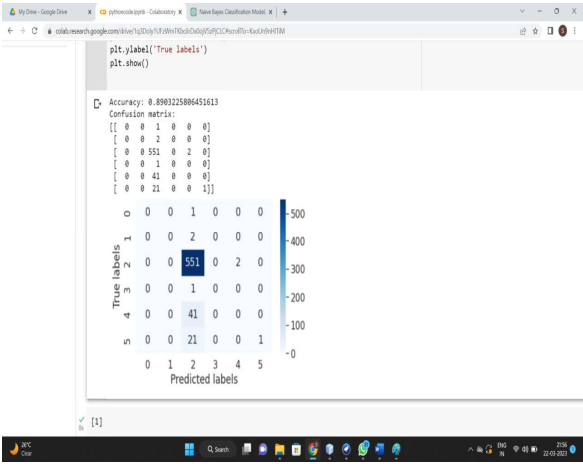


Fig.9 Graphical representation of Naive Bayes Algorithm

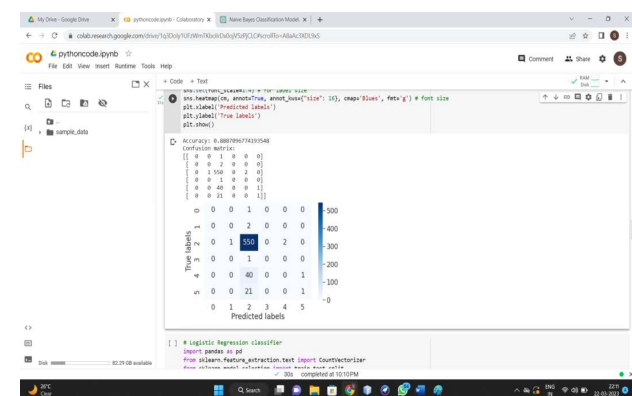


Fig.10 Gradient Boosting Algorithm

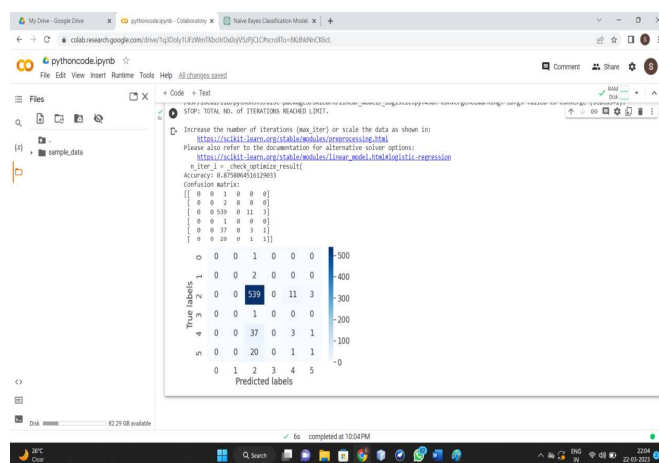


Fig.13 Graphical representation of Logistic Regression Algorithm

VII. CONCLUSION

In this paper the model is developed to forecast court decisions. Machine learning algorithms were applied such as Naïve Bayes, Gradient Boosting and Logistic algorithm were applied and the accuracy has been calculated. The accuracy of Naïve Bayes is 89% and the accuracy of Gradient boosting is 88% and the accuracy of Logistic Regression is 87%. Thus, the Naïve Bayes algorithm gives the highest accuracy.

S. No	Algorithms	Accuracy
1	Naïve Bayes	89%
2	Gradient Boosting	88%
3	Logistic Regression	87%

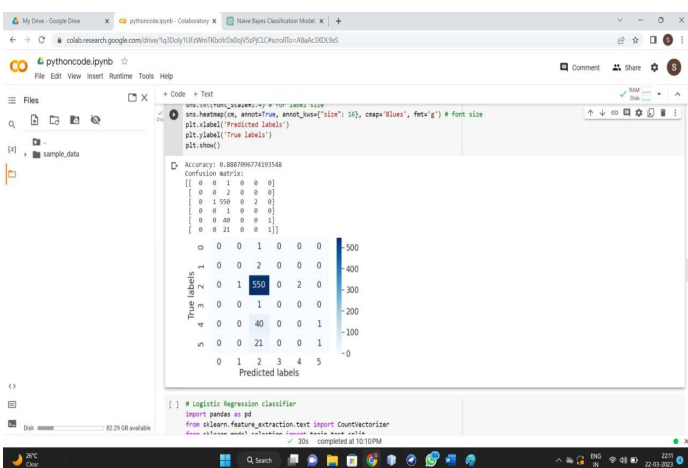


Fig.11 Graphical representation of Gradient Boosting Algorithm

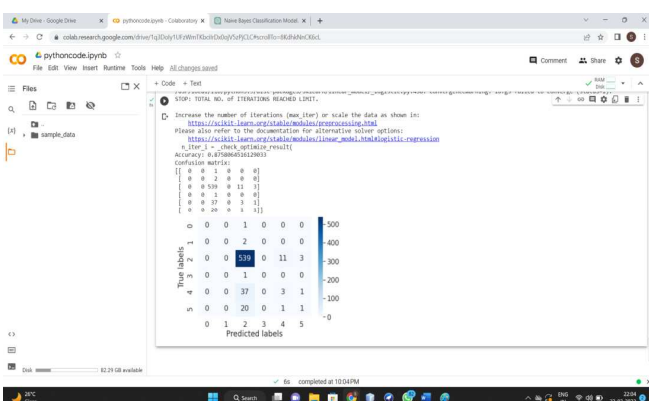


Fig.12 Logistic Regression Classifier

Table 2. Performance of Classifier

As we conclude here that the existing method has a very less accuracy and thus, we increased the accuracy by creating the dataset on our own.

As a result, the system will be easy, enable people to make credible decisions through their legal processes, and engage the public through the legislative in order to improve society.

VIII. FUTURE WORK

In the future, the work will focus on the following. The model has enough accuracy improvements and consistent efforts to achieve this. We also intend to expand our case further evaluation of the dataset and investigation of other effective classifiers with the aim of obtaining better accuracy.

In addition, we strive to provide reasonable end users explanation and predicted outcome. A web-based graphical

user interface that helps end users download the new case smoothly document as input for decision prediction. A working prototype goes even further has evolved into an effective product and aims to be marketed to us project.

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